

IT Job Market Analysis

(Simulated Data) - Project Documentation

Author	Santosh Chaurasia
Project type	Practice project
Tools	Python, SQL (SQL Server), Excel, Power BI
Dataset	Simulated data, 26,500 IT job rows, 2024 to 2026

Note: The data in this project is simulated. I made it with Python using rules that I wrote myself. It is not real hiring data, so the results do not show the real job market. I made this project only to practice the full analysis steps.

1. About the Project

In this project I wanted to practice a full data analysis from start to end. I made a dataset of 26,500 IT job postings from seven Indian cities and looked at skills, salaries, experience levels, cities and companies.

I used four tools: Python to make and prepare the data, SQL Server to answer questions, Excel for pivot tables and a dashboard, and Power BI for a second dashboard.

2. Goals

- Make a jobs table and a skills table and prepare them for SQL.
- Answer questions with SQL: top skills, salary by city, salary tiers, experience vs salary and top companies.
- Explore the data in Python with simple charts.
- Make a dashboard in Excel with pivot tables, charts and slicers.
- Make a dashboard in Power BI with KPI cards, charts and a slicer.

3. Tools Used

Tool	Used for
Python (Pandas)	Making the data, checking it and splitting skills
Plotly	Two simple charts in the analysis notebook
SQL Server	8 queries with JOIN, GROUP BY, CASE, CAST and a subquery
Excel	3 pivot tables, 3 charts, 2 slicers and a dashboard sheet
Power BI	2 KPI cards, 3 charts and an Experience Level slicer

4. Dataset

I made three CSV files.

File	Rows	What it has
simulated_it_jobs.csv	26,500	Main jobs table with 8 columns
Normalized_Job_Skills.csv	127,180	One row for each job and skill (Job_ID, Skill)
IT_Job_Market_Clean.csv	26,500	4 columns (Location, Job_Title, Experience_Level, Salary_INR) used for Excel, Power BI and Python charts

Columns of the main table:

Column	Meaning
Job_ID	Unique ID (JOB_100000 to JOB_126499)
Company_Name	12 companies (TCS, Infosys, Wipro, Cognizant, Accenture, Tech Mahindra, Capgemini, JPMorgan, Amazon India, Allied Digital, eClerx, Parascadd)
Job_Title	8 roles: AI / ML Engineer, Data Analyst, Data Engineer, Java Developer, Python Developer, UI Developer, Manual Tester, System Administrator
Skills_Required	Skills for the role, separated by commas (4 or 5 skills)
Experience_Level	Entry-Level, Mid-Level or Senior
Salary_INR	Salary per year in rupees
Location	Bengaluru, Navi Mumbai, Mumbai, Pune, Hyderabad, Noida, Chennai
Posted_Year	2024, 2025 or 2026

How I made the data: I used a loop in Python that makes 26,500 rows with random choices. These are the rules I used:

- Company, city, experience level and year are picked randomly.
- AI/Data roles (Data Analyst, AI / ML Engineer, Data Engineer) are 30% of jobs in 2024, 50% in 2025 and 70% in 2026. The other jobs are traditional roles.
- Each experience level has a salary range, higher for AI/Data roles. Entry: 3.5-6 LPA (AI/Data) and 2.4-4 LPA (others). Mid: 7-12 LPA and 5-8.5 LPA. Senior: 14-22 LPA and 9.5-15 LPA.
- AI/Data salaries get a 15% increase in 2026.
- Every role has a fixed list of skills.

Because I set these rules myself, many results (like equal jobs in every city) only show the rules and are not real insights.

5. Steps I Followed

- Made the 26,500 job rows in Python and saved them as a CSV file.
- Checked the data with head(), shape, info() and describe(). There are no missing values and no duplicate Job_IDs.
- Split the skills so that each skill has its own row, and saved the skills file.
- Loaded both files in SQL Server (tables it_jobs_main and job_skills_normalized) and ran 8 queries.
- Saved the 4-column clean file with a SQL query.
- Made two charts in Python, then the Excel and Power BI dashboards.

6. SQL

I ran the queries on the database IT_Job_Market_DB. I checked the results with Pandas and they match.

#	Query	Result
1	Top 10 Skills in Demand	Python (15,861) and SQL (14,042) are the top skills. Machine Learning, LLM, Prompt Engineering and NLP have 4,469 each.
2	Top Skills for Data Analyst (JOIN)	SQL, Python, Power BI, Advanced Excel and Data Analysis all have 4,381. It is a tie because every Data Analyst job has the same skills.
3	City-wise Salary for Data Analyst	Mumbai has the highest average (11,69,145 INR) and Navi Mumbai the lowest (11,13,025 INR), about 5% difference.
4	Salary Categories (CASE)	Mid pay (6 to 12 LPA): 9,754 jobs. Entry pay (below 6 LPA): 9,388 jobs. High pay (above 12 LPA): 7,358 jobs.
5	Experience Level vs Salary	Senior 15,67,538 INR, Mid-Level 8,48,152 INR, Entry-Level 4,13,576 INR. Senior pays about 3.8 times Entry.
6	Top 10 Hiring Companies	Cognizant (2,273), Infosys (2,267) and TCS (2,256) are first, but all 12 companies have 2,134 to 2,273 jobs.
7	Top Skills for Python Developer (JOIN)	Python, Django, Flask, SQL and REST APIs have 2,705 each (a tie).
8	Jobs by City, Market Share % (subquery)	Bengaluru has the most (14.60%) and Pune the least (14.00%).

I also wrote one extra query (Query IT_Job_Market_Clean.csv.sql) that selects 4 columns and gives the clean file.

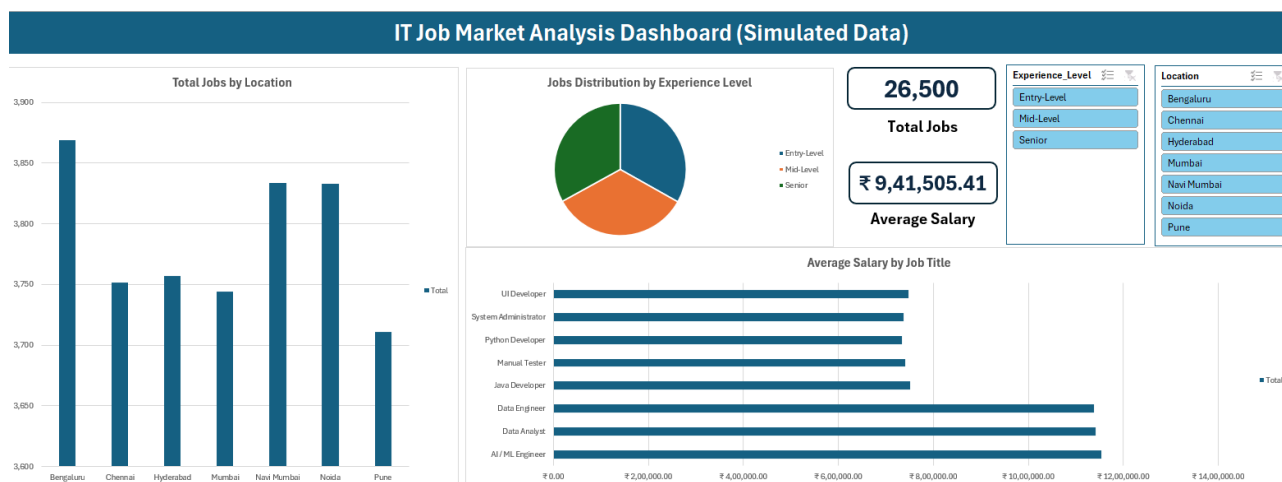
7. Python

Data_Generation_&_Cleaning.ipynb: makes the 26,500 rows, checks the data and makes the skills table. The salary goes from 2.40 LPA to 25.29 LPA, with a mean of 9.42 LPA and a median of 8.09 LPA.

IT_Job_Market_Analysis.ipynb: loads IT_Job_Market_Clean.csv and makes two Plotly charts: average salary by job title and total jobs by location.

8. Excel Dashboard

The workbook IT_Job_Market_Workbook.xlsx has 3 sheets: the data, the pivot tables (jobs by location, average salary by job title and jobs by experience level) and the dashboard. The dashboard has 2 KPI boxes, 3 charts, and slicers for Experience Level and Location. The title says it is simulated data.



Data: simulated dataset generated with Python for practice | Created by Santosh Chaurasia

Excel dashboard (simulated data).

9. Power BI Dashboard

The Power BI dashboard has 2 KPI cards (Total Jobs and Average Salary), a column chart for jobs by location, a pie chart for jobs by experience level, a bar chart for average salary by job title and an Experience Level slicer. The title and the footer say it is simulated data. The cards show rounded numbers (27K and 941.51K) but the real values are 26,500 jobs and 9,41,505 INR.



Data: simulated dataset generated with Python for practice | Created by Santosh Chaurasia

Power BI dashboard (simulated data).

10. What I Found

These results are from the simulated data only.

- There are 26,500 jobs and the average salary is 9.42 LPA (median 8.09 LPA).
- AI/Data roles pay 11.44 LPA on average and other roles pay 7.42 LPA.
- Entry-Level pays 4.14 LPA, Mid-Level 8.48 LPA and Senior 15.68 LPA.
- 35.4% of jobs are below 6 LPA, 36.8% are between 6 and 12 LPA and 27.8% are above 12 LPA.

- The share of AI/Data jobs goes from 30.2% (2024) to 49.5% (2025) to 69.5% (2026).
- AI / ML Engineer, Data Analyst and Data Engineer have the highest average salary (about 11.4 to 11.5 LPA).
- Every city has 14.0% to 14.6% of the jobs, and every company has 2,134 to 2,273 jobs.

11. Limitations

- The data is simulated. Most results come from the random rules I used, so they are not real insights.
- Every role has a fixed skill list, so the skill queries give ties.
- The dashboards use only the 4-column clean file, so they do not show company, skills or year.
- I did not use a random seed, so running the generator again gives slightly different numbers.
- Salary is one yearly number. There is no bonus or other pay.

12. What I Learned

- How to make a dataset with Python and split a column into a separate table.
- How to use JOIN, CASE, subquery and GROUP BY in SQL Server.
- How to make pivot tables, charts and slicers in Excel, and a dashboard in Power BI.
- That results from simulated data mostly show the rules I put in it.

13. How to Run

- **Python:** open the notebooks in Jupyter or Colab (pip install pandas plotly). Running the generator makes a new random dataset, so use the CSV files in the Data folder to get the same numbers.
- **SQL:** run `create_db.sql`, import `simulated_it_jobs.csv` as `it_jobs_main` and `Normalized_Job_Skills.csv` as `job_skills_normalized`, then run the query files.
- **Excel and Power BI:** open the files in the Excel and Power BI folders.

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